

## LAHR, Germany

WMO: 108050

Lat: 48.3647N

Lon: 7.8281E

Elev: 157

StdP: 99.45

Time Zone: 1.00 (EUC)

Period: 95-13

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -8.9       | -6.4 | -11.6                         | 1.4 | -7.8 | -9.1 | 1.8 | -4.8 | 19.7                  | 10.3 | 14.2 | 7.9  | 2.0                   | 20   | 0.480 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.7                   | 31.0            | 20.8 | 29.1 | 20.2 | 27.3 | 19.5 | 21.8                | 28.6 | 21.0 | 27.3 | 20.2 | 25.9 | 2.4                  | 210  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 19.7                            | 14.7 | 24.4 | 18.9 | 14.0 | 23.5 | 18.1 | 13.3 | 22.6 | 64.4          | 28.7 | 61.4 | 27.4 | 58.7 | 25.9 | 27.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 9.1               | 7.4  | 6.3 | -12.6                      | 34.3 | 2.8                | 1.8 | -14.7                                              | 35.6 | -16.3      | 36.7 | -17.9      | 37.7 | -20.0      | 39.0 |
|                   |      |     | -13.0                      | 23.9 | 2.8                | 1.1 | -15.0                                              | 24.6 | -16.7      | 25.2 | -18.2      | 25.8 | -20.3      | 26.6 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 11.0   | 1.9   | 3.4   | 6.8   | 10.8  | 15.1  | 18.4  | 19.8  | 19.7  | 15.4  | 11.3  | 6.0   | 2.8  |
|                                                                                  | DBStd       | 7.50   | 4.87  | 4.58  | 4.05  | 3.72  | 3.54  | 3.46  | 2.98  | 2.98  | 3.17  | 3.89  | 3.93  | 4.58 |
|                                                                                  | HDD10.0     | 978    | 253   | 187   | 113   | 35    | 3     | 0     | 0     | 0     | 1     | 32    | 129   | 227  |
|                                                                                  | HDD18.3     | 2872   | 510   | 418   | 357   | 227   | 111   | 43    | 18    | 19    | 97    | 219   | 371   | 483  |
|                                                                                  | CDD10.0     | 1340   | 2     | 2     | 14    | 58    | 161   | 251   | 305   | 301   | 164   | 72    | 8     | 2    |
|                                                                                  | CDD18.3     | 192    | 0     | 0     | 0     | 0     | 11    | 44    | 64    | 61    | 10    | 1     | 0     | 0    |
|                                                                                  | CDH23.3     | 1772   | 0     | 0     | 0     | 19    | 137   | 404   | 578   | 533   | 92    | 8     | 0     | 0    |
|                                                                                  | CDH26.7     | 522    | 0     | 0     | 0     | 1     | 27    | 114   | 189   | 177   | 14    | 1     | 0     | 0    |
| Wind                                                                             | WSAvg       | 2.5    | 3.0   | 3.2   | 2.8   | 2.7   | 2.6   | 2.3   | 2.2   | 2.1   | 2.1   | 2.2   | 2.3   | 2.7  |
| Precipitation                                                                    | PrecAvg     | 895    | 54    | 55    | 59    | 65    | 91    | 92    | 99    | 82    | 74    | 79    | 75    | 70   |
|                                                                                  | PrecMax     | 1121   | 123   | 103   | 169   | 129   | 165   | 179   | 289   | 186   | 130   | 168   | 157   | 145  |
|                                                                                  | PrecMin     | 638    | 9     | 16    | 12    | 5     | 24    | 29    | 29    | 13    | 28    | 7     | 2     | 4    |
|                                                                                  | PrecStd     | 114    | 30    | 25    | 36    | 32    | 38    | 37    | 50    | 37    | 30    | 36    | 36    | 35   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 13.9  | 15.7  | 20.7  | 25.6  | 29.7  | 32.5  | 33.4  | 34.8  | 28.7  | 24.2  | 17.2  | 14.4 |
|                                                                                  |             | MCWB   | 10.7  | 9.6   | 11.9  | 15.3  | 18.9  | 21.0  | 21.2  | 21.7  | 19.3  | 17.6  | 12.8  | 10.4 |
|                                                                                  | 2%          | DB     | 12.3  | 13.8  | 18.1  | 22.9  | 26.7  | 29.8  | 30.9  | 30.4  | 25.9  | 21.1  | 14.8  | 12.3 |
|                                                                                  |             | MCWB   | 9.7   | 9.3   | 11.4  | 13.9  | 17.7  | 20.5  | 20.8  | 21.2  | 19.1  | 16.0  | 11.7  | 9.8  |
|                                                                                  | 5%          | DB     | 11.0  | 11.9  | 15.5  | 20.3  | 24.6  | 27.7  | 28.8  | 28.4  | 23.8  | 19.1  | 13.1  | 10.8 |
|                                                                                  |             | MCWB   | 8.6   | 8.7   | 10.5  | 13.3  | 16.6  | 19.9  | 20.2  | 20.2  | 17.7  | 15.2  | 10.8  | 8.8  |
|                                                                                  | 10%         | DB     | 9.1   | 10.0  | 13.3  | 18.1  | 22.4  | 25.7  | 26.7  | 26.3  | 21.8  | 17.1  | 11.6  | 9.2  |
|                                                                                  |             | MCWB   | 7.2   | 7.4   | 9.7   | 12.2  | 15.8  | 18.9  | 19.3  | 19.5  | 16.7  | 14.1  | 9.7   | 7.7  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 11.5  | 11.0  | 13.5  | 16.3  | 20.7  | 22.7  | 22.7  | 23.2  | 21.0  | 17.9  | 13.8  | 11.6 |
|                                                                                  |             | MCDB   | 13.3  | 14.3  | 18.2  | 23.4  | 26.7  | 29.5  | 30.0  | 30.7  | 26.3  | 22.3  | 15.7  | 13.0 |
|                                                                                  | 2%          | WB     | 10.1  | 9.8   | 12.2  | 14.8  | 18.9  | 21.4  | 21.6  | 21.8  | 19.5  | 16.7  | 12.3  | 10.1 |
|                                                                                  |             | MCDB   | 11.9  | 13.0  | 15.8  | 21.2  | 25.0  | 27.5  | 28.8  | 28.6  | 24.4  | 20.1  | 14.2  | 11.9 |
|                                                                                  | 5%          | WB     | 8.6   | 8.7   | 11.2  | 13.7  | 17.5  | 20.4  | 20.7  | 20.9  | 18.4  | 15.6  | 11.1  | 9.0  |
|                                                                                  |             | MCDB   | 10.7  | 11.5  | 14.7  | 19.3  | 22.9  | 26.4  | 27.2  | 26.9  | 22.5  | 18.4  | 12.8  | 10.6 |
|                                                                                  | 10%         | WB     | 7.3   | 7.5   | 10.0  | 12.6  | 16.3  | 19.4  | 19.8  | 20.0  | 17.3  | 14.5  | 9.9   | 7.8  |
|                                                                                  |             | MCDB   | 9.0   | 9.8   | 13.4  | 17.1  | 21.0  | 24.7  | 25.3  | 25.3  | 20.8  | 16.6  | 11.3  | 9.2  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 5.4   | 7.1   | 9.1   | 10.8  | 10.5  | 10.8  | 10.7  | 10.8  | 10.6  | 8.7   | 6.2   | 5.0  |
|                                                                                  |             | MCDBR  | 7.0   | 10.1  | 13.4  | 15.3  | 15.1  | 14.2  | 14.6  | 15.0  | 14.0  | 11.5  | 8.3   | 6.3  |
|                                                                                  |             | MCWBR  | 5.1   | 6.6   | 7.9   | 7.9   | 7.1   | 6.3   | 6.1   | 6.6   | 7.4   | 6.8   | 5.7   | 4.8  |
|                                                                                  | 5% WB       | MCDBR  | 6.4   | 8.7   | 11.6  | 13.7  | 13.1  | 13.2  | 13.0  | 13.5  | 12.1  | 10.4  | 7.7   | 6.1  |
|                                                                                  |             | MCWBR  | 5.1   | 5.9   | 7.2   | 7.3   | 6.7   | 6.1   | 5.8   | 6.4   | 6.7   | 6.5   | 5.8   | 4.8  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.314  | 0.337 | 0.368 | 0.403 | 0.413 | 0.420 | 0.417 | 0.413 | 0.388 | 0.365 | 0.334 | 0.310 |      |
|                                                                                  | taud        | 2.456  | 2.386 | 2.329 | 2.255 | 2.268 | 2.285 | 2.284 | 2.315 | 2.361 | 2.416 | 2.458 | 2.454 |      |
|                                                                                  | Ebn at Noon | 757    | 814   | 843   | 845   | 850   | 845   | 841   | 827   | 812   | 770   | 723   | 710   |      |
|                                                                                  | Edh at Noon | 64     | 86    | 106   | 126   | 129   | 128   | 126   | 117   | 102   | 83    | 64    | 58    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.96   | 1.72  | 2.94  | 4.28  | 5.05  | 5.82  | 5.44  | 4.74  | 3.58  | 2.16  | 1.11  | 0.77  |      |
|                                                                                  | RadStd      | 0.14   | 0.29  | 0.45  | 0.60  | 0.55  | 0.48  | 0.51  | 0.48  | 0.37  | 0.19  | 0.12  | 0.14  |      |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (91 neighbors) |  | N/A   | N/A     | N/A    | N/A     | +0.36 | +0.30 | N/A         | N/A     | +51     | N/A     |

Nomenclature: See separate page